

The empirical recurrence for OEIS A206537, proved by transfer matrix

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Abstract

OEIS A206537 counts arrays of width 4 over $\{0, 1, 2\}$ subject to a condition read round the perimeter of every 2×2 subblock: the clockwise counts increase rightwards and downwards, the counterclockwise counts decrease. The entry carries an empirical recurrence of order 33 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: the condition is settled inside a bounded window of consecutive rows, so the count is a walk count on a finite digraph with $S = 412$ vertices, and whether a linear recurrence holds for such a count is a finite computation.

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1 The conjecture

OEIS A206537 is “Number of $(n+1) \times 4$ 0..2 arrays with the number of clockwise edge increases in 2×2 subblocks nondecreasing, and counterclockwise edge increases nonincreasing, rightwards and downwards.”. It has offset 1 and begins

1620, 14781, 141759, 1419921, 14528199, 150387285, 1567749888, 16417626507, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 26*a(n-1) - 169*a(n-2) - 962*a(n-3) + 15074*a(n-4) - 23548*a(n-5) - 388835*a(n-6) + 1661306*a(n-7) + 3387121*a(n-8) - 32428954*a(n-9) + 18217134*a(n-10) + 298256916*a(n-11) - 598690104*a(n-12) - 1238898284*a(n-13) + 4947375888*a(n-14) + 248363732*a(n-15) - 19612727472*a(n-16) + 17680860320*a(n-17) + 37546981780*a(n-18) - 68886915424*a(n-19) - 19805672744*a(n-20) + 117856544624*a(n-21) - 46294103104*a(n-22) - 94823075520*a(n-23) + 86888550208*a(n-24) + 22051127568*a(n-25) - 55468199168*a(n-26) + 13452820320*a(n-27) + 12246999328*a(n-28) - 7344098112*a(n-29) + 380505984*a(n-30) + 639121152*a(n-31) - 167178240*a(n-32) + 11612160*a(n-33)$ for $n > 37$.

As of the “Last modified” line on the live entry (Dec 13 2015 16:32:02, revision 7) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The arrays have 4 columns and a growing number of rows, with entries in $\{0, 1, 2\}$. A 2×2 subblock is a set of four cells

$$B = \begin{pmatrix} p & q \\ r & s \end{pmatrix}, \quad p = x(i, j), \quad q = x(i, j + 1), \quad r = x(i + 1, j), \quad s = x(i + 1, j + 1).$$

Its perimeter is a cycle of four cells, walked clockwise as $p \rightarrow q \rightarrow s \rightarrow r \rightarrow p$ and counterclockwise as $p \rightarrow r \rightarrow s \rightarrow q \rightarrow p$. Write

$$\text{cw}(B) = [p < q] + [q < s] + [s < r] + [r < p], \quad \text{ccw}(B) = [p < r] + [r < s] + [s < q] + [q < p]$$

for the number of steps at which the value rises, going each way round.

Lemma 1. $\text{cw}(B) + \text{ccw}(B) = 4 - \#\{\text{edges of the cycle with equal ends}\}$.

Proof. Each of the four edges of the square is walked once by each cycle, and in opposite directions. An edge with distinct ends therefore contributes exactly one rise, to one of the two counts; an edge with equal ends contributes to neither. $\square$

The condition the entry imposes is this: whenever B' is the subblock one step to the right of B , or one step below it,

$$\text{cw}(B) \leq \text{cw}(B'), \quad \text{ccw}(B) \geq \text{ccw}(B')$$

so the clockwise count is nondecreasing rightwards and downwards while the counterclockwise count is nonincreasing in both directions.

3 A bounded window

Two consecutive rows decide every subblock lying between them, and the conditions of Section 2 compare such a subblock with the subblock directly above it, which lies between the PREVIOUS two rows. So the state carries the row just written together with the clockwise and counterclockwise counts of each subblock of that pair, one small number per column: at most five values each, whatever the alphabet.

One step of the walk writes a whole row. The row is filled left to right, and the subblock closed by positions $j-1, j$ is tested the moment position j is written, so a partial row that cannot be completed is abandoned rather than enumerated. The walk starts at the state standing for the empty array, so that a walk of n steps is an array of n rows.

The end vector is all ones: an array may stop after any row. Thus

$$a(n) = \iota^\top M^j \tau, \quad S = 412,$$

for the transfer matrix M on those S states. The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite, and every question about linear recurrences it satisfies is decidable.

4 The proof

Lemma 2. Let $q(t) = t^{33} - \sum_i c_i t^{33-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.

Proof. Factor M^j out of $M^{j+33} - \sum_i c_i M^{j+33-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 3.

$a(n) = 26a(n-1) - 169a(n-2) - 962a(n-3) + 15074a(n-4) - 23548a(n-5) - 388835a(n-6) + 1661306a(n-7) + \dots$
for every $n > 37$.

Proof. The vector $w = q(M)\tau$ was formed by 33 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 37 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: every array of the stated width and a few rows was written out cell by cell over $\{0, 1, 2\}$, each of its 2×2 subblocks was read round its perimeter, and the condition of Section 2 was applied as stated. No window, no state and no recurrence is involved in that computation, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.